Analytical Simplifications for the Production PNP–BV System

Jake Weinstein

May 3, 2026

Abstract

The current production forward model solves a three-species log-concentration Poisson–Nernst–Planck system with log-rate Butler–Volmer boundary kinetics and an analytical Boltzmann ClO_4^- counterion. A closed-form solution of the full two-dimensional steric PNP–BV system is not realistic, but the model does contain several exploitable analytical structures. The main simplification is a matched-asymptotic reduction: represent the diffuse Debye layer with an analytical Poisson–Boltzmann profile, solve only an outer electroneutral diffusion–reaction problem, and initialize the full solver with the resulting composite state. In the electroneutral outer region, the existing Boltzmann ClO_4^- constraint makes proton transport ambipolar:

$$J_{\mathrm{H}^+} = -2D_{\mathrm{H}^+}\nabla c_{\mathrm{H}^+}.$$

This collapses the one-dimensional steady outer problem to linear profiles coupled to a two-rate Butler–Volmer algebraic system. The resulting voltage-specific initial condition directly targets the positive-voltage Newton failure mode: the solver must start close to the exponentially depleted Boltzmann proton layer rather than discover it from a linear-potential guess.

1 Production model and question

The active production stack is the formulation used by `Forward/bv_solver/forms_logc.py` and `Forward/bv_solver/boltzmann.py`. The dynamic species are

$$[\mathrm{O_2}, \mathrm{H_2O_2}, \mathrm{H^+}], \qquad z = [0, 0, +1],$$

while inert ClO_4^- is removed as a Nernst–Planck unknown and enters Poisson's equation analytically as

$$c_-(x) = c_-^b \exp(\phi(x)), \tag{1}$$

where $\phi = \Phi/V_T$ is the nondimensional electrostatic potential. The remaining concentrations are represented by

$$u_i = \log c_i, \qquad c_i = \exp(u_i).$$

The electrode reactions are

$$R_1: \quad \mathrm{O_2} + 2\mathrm{H^+} + 2e^- \rightleftharpoons \mathrm{H_2O_2}, \qquad E_{\mathrm{eq}}^{(1)} = 0.68 \text{ V vs RHE}, \tag{2}$$

$$R_2: \quad \mathrm{H_2O_2} + 2\mathrm{H^+} + 2e^- \to 2\mathrm{H_2O}, \qquad E_{\mathrm{eq}}^{(2)} = 1.78 \text{ V vs RHE}. \tag{3}$$

The question is whether this PNP–BV system admits analytical simplifications similar to the Boltzmann counterion reduction, whether the whole system can be solved analytically, and whether a useful analytical initial condition can be constructed for a specified voltage and experimental setup.

2 Main conclusion

A closed-form solution of the full production system should not be expected. The full problem is nonlinear, singularly perturbed, steric, and two-dimensional, with finite reaction fluxes coupled through nonlinear Butler–Volmer boundary rates. Even a one-dimensional steady PNP–BV problem with nonzero reactive flux generally does not reduce to elementary closed form.

The useful analytical object is instead a matched-asymptotic composite state:

1. Use Poisson–Boltzmann or Gouy–Chapman theory for the charged Debye layer adjacent to the electrode.
2. Use an electroneutral outer problem away from that layer.
3. In the outer region, use the analytic ClO_4^- relation to collapse proton electromigration into ambipolar diffusion.
4. Couple the resulting linear one-dimensional outer profiles to a small Butler–Volmer algebraic system for R_1 and R_2.

This is not a replacement for the full PNP–BV solve. It is a high-quality analytical predictor and initial condition, especially at positive voltages where the current linear initial potential profile is far from the physical Boltzmann layer.

3 Exact zero-flux simplification

Ignoring steric terms for the moment, the log-concentration Nernst–Planck flux can be written as

$$J_i = -D_i c_i \nabla(u_i + z_i \phi). \tag{4}$$

Define the electrochemical-potential-like variable

$$\mu_i = u_i + z_i \phi. \tag{5}$$

Then

$$J_i = -D_i c_i \nabla \mu_i. \tag{6}$$

Therefore any zero-flux species in a connected region satisfies

$$\nabla \mu_i = 0, \qquad c_i = C_i \exp(-z_i \phi).$$

Equation (1) is exactly this simplification applied to the inert supporting electrolyte counterion. The same argument applies locally to H^+ inside the diffuse layer whenever the normal reaction flux is small relative to rapid double-layer equilibration:

$$c_{\mathrm{H}^+}(y) \approx c_{\mathrm{H}^+}^{\mathrm{outer}} \exp[-(\phi(y) - \phi_{\mathrm{outer}})]. \tag{7}$$

This is not globally exact because protons are consumed by both reactions, but it is precisely the approximation needed in the thin region that currently causes Newton failures.

With steric effects active, the zero-flux potential is modified to

$$\mu_i = u_i + z_i \phi + \mu_{\mathrm{steric}}(c).$$

The Boltzmann factor is then modified by an activity coefficient. For the current dilute production cases with $a = 0.01$, the nonsteric formula is the right first analytical initial condition. A steric Picard correction can be added after the nonsteric profile works.

4 Analytical Debye layer

Let h_o be the outer proton concentration at the edge of the diffuse layer. Electroneutrality with the analytic ClO_4^- ion gives the outer potential

$$\phi_o = \log\left(\frac{h_o}{c_-^b}\right). \tag{8}$$

Define the diffuse-layer potential drop

$$\psi(y) = \phi(y) - \phi_o. \tag{9}$$

For a symmetric monovalent H^+/ClO_4^- layer, the local Boltzmann profiles are

$$c_{\mathrm{H}^+} = h_o \exp(-\psi), \qquad c_- = h_o \exp(+\psi). \tag{10}$$

The one-dimensional inner Poisson equation becomes

$$\varepsilon\, \psi'' = 2h_o \sinh\psi, \qquad \lambda_{D,o} = \sqrt{\frac{\varepsilon}{2h_o}}, \tag{11}$$

where ε is the nondimensional Poisson coefficient $(\lambda_D/L)^2$ in the code.

On a semi-infinite domain, the Gouy–Chapman solution is

$$\tanh\left(\frac{\psi(y)}{4}\right) = \tanh\left(\frac{\psi_D}{4}\right)\exp\left(-\frac{y}{\lambda_{D,o}}\right), \qquad \psi_D = \phi_e - \phi_o, \tag{12}$$

or equivalently

$$\psi(y) = 4\ \mathrm{atanh}\left[\tanh\left(\frac{\psi_D}{4}\right)\exp\left(-\frac{y}{\lambda_{D,o}}\right)\right]. \tag{13}$$

For small voltage, the Debye–Huckel approximation is enough:

$$\psi(y) \approx \psi_D \exp(-y/\lambda_{D,o}), \qquad c_{\mathrm{H}^+} \approx h_o(1-\psi), \qquad c_- \approx h_o(1+\psi).$$

For the production voltage grid, especially on the positive side, the full Gouy–Chapman expression is the better default.

5 Combined Boltzmann–Tafel factor

Both production reactions contain the cathodic proton factor $(c_{\mathrm{H}^+}/c_{\mathrm{H}^+}^{\mathrm{ref}})^2$. In the diffuse-layer approximation,

$$c_{\mathrm{H}^+,s} = h_o \exp(-\psi_D).$$

Thus each cathodic Butler–Volmer branch contains

$$\left(\frac{c_{\mathrm{H}^+,s}}{c_{\mathrm{H}^+}^{\mathrm{ref}}}\right)^2 \exp(-\alpha_j n\eta_j) = \left(\frac{h_o}{c_{\mathrm{H}^+}^{\mathrm{ref}}}\right)^2 \exp(-2\psi_D - \alpha_j n\eta_j). \tag{14}$$

This compactly explains the positive-voltage conditioning wall. At anodic $\psi_D \gg 1$, the cathodic branches are suppressed not only by the usual Tafel factor, but also by two-proton Boltzmann depletion. At $V_{\mathrm{RHE}} = +1.0$ V, ψ_D can be $O(40)$, so a physically reasonable initial condition has u_{H^+} tens of log-units below bulk at the electrode.

6 Outer electroneutral simplification

Outside the Debye layer, electroneutrality gives

$$c_{\mathrm{H}^+} \approx c_-^b \exp(\phi), \qquad \phi = \log\left(\frac{c_{\mathrm{H}^+}}{c_-^b}\right). \tag{15}$$

Insert this relation into the proton Nernst–Planck flux:

$$\begin{aligned} J_{\mathrm{H}^+} &= -D_{\mathrm{H}^+} c_{\mathrm{H}^+} \nabla(\log c_{\mathrm{H}^+} + \phi) && (16) \\ &= -D_{\mathrm{H}^+} c_{\mathrm{H}^+} \nabla\left(\log c_{\mathrm{H}^+} + \log \frac{c_{\mathrm{H}^+}}{c_-^b}\right) && (17) \\ &= -2D_{\mathrm{H}^+} \nabla c_{\mathrm{H}^+}. && (18) \end{aligned}$$

Therefore the outer proton equation is ordinary diffusion with effective diffusivity $2D_{\mathrm{H}^+}$. The neutral species are already ordinary diffusion:

$$J_{\mathrm{O}_2} = -D_{\mathrm{O}_2} \nabla c_{\mathrm{O}_2}, \qquad J_{\mathrm{H}_2\mathrm{O}_2} = -D_{\mathrm{H}_2\mathrm{O}_2} \nabla c_{\mathrm{H}_2\mathrm{O}_2}.$$

In a one-dimensional steady outer approximation, all three outer concentrations are linear functions of distance from the electrode.

7 Two-rate outer algebraic model

Let $y = 0$ be the electrode and $y = 1$ the bulk. Let R_1 and R_2 be positive in the cathodic directions used by the implementation:

$$R_1 : \mathrm{O}_2 \to \mathrm{H}_2\mathrm{O}_2, \qquad R_2 : \mathrm{H}_2\mathrm{O}_2 \to \mathrm{H}_2\mathrm{O}.$$

The production stoichiometries are

$$s_{R_1} = [-1, +1, -2], \qquad s_{R_2} = [0, -1, -2].$$

For the linear outer profiles, the electrode-side outer concentrations are

$$\begin{aligned} O_s &= O_b - \frac{R_1}{D_{\mathrm{O}_2}}, && (19) \\ P_s &= P_b + \frac{R_1 - R_2}{D_{\mathrm{H}_2\mathrm{O}_2}}, && (20) \\ h_o &= H_b - \frac{R_1 + R_2}{D_{\mathrm{H}^+}}. && (21) \end{aligned}$$

The proton expression uses the ambipolar diffusivity $2D_{\mathrm{H}^+}$ and the two-proton stoichiometry, so the factors cancel.

The proton concentration entering the BV rate is the outer proton value multiplied by the inner Boltzmann factor:

$$H_s = h_o \exp(-\psi_D), \qquad \psi_D = \phi_e - \phi_o, \qquad \phi_o = \log(h_o/c_-^b). \tag{22}$$

For the current no-Stern production stack, the code evaluates

$$\eta_j = \phi_{\text{applied}} - E_{\text{eq}}^{(j)}$$

in the BV exponent. If a Stern/Frumkin formulation is enabled, this should be replaced by that formulation's corrected $\eta_j = \phi_m - \phi_s - E_{\mathrm{eq}}^{(j)}$.

Holding H_s fixed for one Picard step, define

$$A_1 = k_1 \left(\frac{H_s}{H_{\mathrm{ref}}}\right)^2 \exp(-\alpha_1 n \eta_1), \qquad B_1 = k_1 \exp((1-\alpha_1) n \eta_1), \tag{23}$$

$$A_2 = k_2 \left(\frac{H_s}{H_{\mathrm{ref}}}\right)^2 \exp(-\alpha_2 n \eta_2). \tag{24}$$

The reduced BV rates are

$$R_1 = A_1 O_s - B_1 P_s, \qquad R_2 = A_2 P_s.$$

Substituting (21) gives a 2×2 linear system:

$$\begin{bmatrix} 1 + A_1/D_{\mathrm{O_2}} + B_1/D_{\mathrm{H_2O_2}} & -B_1/D_{\mathrm{H_2O_2}} \\ -A_2/D_{\mathrm{H_2O_2}} & 1 + A_2/D_{\mathrm{H_2O_2}} \end{bmatrix} \begin{bmatrix} R_1 \\ R_2 \end{bmatrix} = \begin{bmatrix} A_1 O_b - B_1 P_b \\ A_2 P_b \end{bmatrix}. \tag{25}$$

This algebraic solve gives a voltage-specific approximation to surface rates and concentrations. Updating h_o, ϕ_o, ψ_D, and the coefficients A_1, A_2, B_1 by fixed point yields a cheap nonlinear reduced model.

Useful limiting cases follow immediately:

- If R_1 is one-way cathodic and H_s is fixed, then

  $$R_1 = \frac{A_1 O_b}{1 + A_1/D_{\mathrm{O_2}}},$$

  recovering the transition from kinetic control to the O_2 diffusion limit.
- If R_2 is one-way and R_1 is already estimated, then

  $$R_2 = \frac{A_2(P_b + R_1/D_{\mathrm{H_2O_2}})}{1 + A_2/D_{\mathrm{H_2O_2}}}.$$

- If anodic R_1 dominates, the same algebraic system enforces peroxide depletion through P_s; the rate cannot exceed the peroxide supply implicit in $P_s \geq 0$.

8 Voltage-specific analytical initial condition

The most practical output is an initial condition for the full log-c PNP–BV solver. For a specified voltage and experiment:

1. Convert voltage to nondimensional potential $\phi_e = V_{\mathrm{RHE}}/V_T$.
2. Initialize $R_1 = R_2 = 0$, $h_o = H_b$, $\phi_o = 0$, and $\psi_D = \phi_e$.
3. Compute $H_s = h_o \exp(-\psi_D)$.
4. Build A_1, B_1, and A_2 from the BV parameters.
5. Solve (25) for R_1, R_2.

6. Update
$$O_s = O_b - \frac{R_1}{D_{\mathrm{O_2}}}, \quad P_s = P_b + \frac{R_1 - R_2}{D_{\mathrm{H_2O_2}}}, \quad h_o = H_b - \frac{R_1 + R_2}{D_{\mathrm{H^+}}},$$
and then
$$\phi_o = \log(h_o/c_-^b), \qquad \psi_D = \phi_e - \phi_o.$$
7. Under-relax and repeat until the rates stop changing.
8. Construct outer linear profiles:
$$O_{\mathrm{out}}(y) = O_s + (O_b - O_s)y, \tag{26}$$
$$P_{\mathrm{out}}(y) = P_s + (P_b - P_s)y, \tag{27}$$
$$H_{\mathrm{out}}(y) = h_o + (H_b - h_o)y. \tag{28}$$
9. Construct the Debye correction:
$$\lambda_{D,o} = \sqrt{\frac{\varepsilon}{2\max(h_o, h_{\min})}},$$
$$\psi(y) = 4\ \mathrm{atanh}\left[\tanh\left(\frac{\psi_D}{4}\right)\exp\left(-\frac{y}{\lambda_{D,o}}\right)\right].$$
10. Build the composite initial condition:
$$\phi_{\mathrm{IC}}(y) = \log\left(\frac{H_{\mathrm{out}}(y)}{c_-^b}\right) + \psi(y), \tag{29}$$
$$H_{\mathrm{IC}}(y) = H_{\mathrm{out}}(y)\exp[-\psi(y)], \tag{30}$$
$$O_{\mathrm{IC}}(y) = O_{\mathrm{out}}(y), \tag{31}$$
$$P_{\mathrm{IC}}(y) = \max(P_{\mathrm{out}}(y), P_{\min}), \tag{32}$$
$$u_{i,\mathrm{IC}}(y) = \log c_{i,\mathrm{IC}}(y). \tag{33}$$

For the two-dimensional rectangle, this is used as a y-only profile and interpolated across x.

This IC places Newton near the correct Debye-layer manifold from the start. It also respects the outer reaction-diffusion mass balances instead of setting every concentration to its bulk value.

9 Can $\mathrm{H^+}$ be removed analytically like $\mathrm{ClO_4^-}$?

Not exactly in the full model. Protons participate in both BV reactions through the $(c_{\mathrm{H^+}}/c_{\mathrm{H^+}}^{\mathrm{ref}})^2$ factors and have nonzero reactive boundary flux. Removing them globally would erase a real transport limitation.

There are three controlled reductions:

1. **Inner-layer Boltzmann $\mathrm{H^+}$.** Use the proton Boltzmann relation only inside the Debye layer. This is the least invasive and most useful initialization strategy.
2. **Outer ambipolar $\mathrm{H^+}$.** Replace outer proton NP transport with $J_{\mathrm{H^+}} = -2D_{\mathrm{H^+}}\nabla c_{\mathrm{H^+}}$. This keeps proton depletion while removing outer Poisson stiffness.
3. **All-Boltzmann $\mathrm{H^+}$ surrogate.** Set $H_s = H_b\exp(-\psi_D)$ directly and drop proton transport. This may be a fast surrogate or diagnostic, but it is a different physical model.

The recommended model-development path is the first two together: an outer electroneutral diffusion–BV solve plus analytical Debye-layer matching.

10 Relationship to Stern-layer support

The current no-Stern production mode imposes Dirichlet $\phi = \phi_{\text{applied}}$ at the electrode. This puts the full applied potential into the resolved diffuse layer, making large positive voltages extremely stiff.

Physically, this is a strong assumption. It identifies the solution-side potential at the reaction plane with the metal potential:

$$\phi_s = \phi_m = \phi_{\text{applied}}.$$

Real electrode/electrolyte interfaces normally contain a compact Stern layer. Ions have a finite closest-approach distance, solvent is structured near the surface, and some voltage can drop across this compact region before the diffuse PNP layer begins. The applied metal-to-bulk voltage is therefore more naturally split as

$$\phi_m - \phi_{\text{bulk}} = \Delta\phi_{\text{Stern}} + \Delta\phi_{\text{Diffuse}}.$$

The current Dirichlet model effectively sets $\Delta\phi_{\text{Stern}} = 0$, so the resolved diffuse layer must absorb the whole interfacial potential drop.

A Stern formulation replaces that electrode Dirichlet condition with a capacitance relation and evaluates the BV overpotential as

$$\eta = \phi_m - \phi_s - E_{\text{eq}}.$$

The capacitance law is

$$\sigma = C_S(\phi_m - \phi_s),$$

where C_S is the Stern-layer capacitance. In the PNP weak form this is a Robin-type electrostatic boundary condition, not a prescribed-potential condition. The comparison is:

Dirichlet:	prescribe the solution potential ϕ_s,
Stern:	prescribe the metal potential ϕ_m and relate diffuse-layer charge to $\phi_m - \phi_s$.

This is generally more physical for an electrochemical interface, but it introduces the additional physical parameter C_S. If C_S is poorly chosen, the model can be structurally more physical while still being less predictive.

The limiting behavior is important:

$$C_S \to \infty \quad \Longrightarrow \quad \phi_m - \phi_s \to 0,$$

which recovers the current Dirichlet-like no-Stern behavior. For finite C_S, ϕ_s is not pinned to ϕ_m; some applied voltage can drop in the compact layer. For the positive-voltage solver problem, this matters because it can keep the resolved solution-side potential ϕ_s much smaller than the metal voltage ϕ_m, reducing the extreme Boltzmann source $c_-^b \exp(\phi_s)$ that currently drives the Poisson residual wall.

The same analytical framework applies, but ψ_D is no longer simply $\phi_e - \phi_o$. It should be solved with a compact capacitance relation:

$$\sigma_{\text{GC}}(\psi_D) = C_S(\phi_m - \phi_s), \qquad \phi_s = \phi_o + \psi_D.$$

The Gouy–Chapman surface charge follows from the first integral,

$$\sigma_{\text{GC}} \sim \text{sign}(\psi_D)\sqrt{8\varepsilon h_o}\,\sinh\left(\frac{|\psi_D|}{2}\right),$$

up to the exact sign convention used by the weak Poisson boundary term. This is more than an initializer: it is a route to replacing the explicit Debye layer with an algebraic boundary condition.

11 What is solved vs. what is analytical

The safest first use is not to replace the model. The analytical Debye/Stern construction should first be used as an initial condition for the existing full PNP–BV solver.

In the current resolved model, the PDE solver computes the outer transport, the explicit Debye layer, and the BV surface flux. In the proposed analytical-initial-condition workflow, the analytical formula computes only a better starting profile for the Debye-layer potential and proton correction; the same full PNP–BV equations are still solved afterward.

A later reduced model would make a stronger split:

PDE solver:	outer O_2, H_2O_2, H^+ transport, and possibly an outer electroneutral potential relation,
Analytical boundary map:	ϕ_s from Stern/diffuse-layer charge balance,
	$H_s = H_o \exp[-(\phi_s - \phi_o)]$,
	$O_s \approx O_o, \quad P_s \approx P_o$,
BV flux law:	O_2 flux $= -R_1$,
	H_2O_2 flux $= R_1 - R_2$,
	H^+ flux $= -2R_1 - 2R_2$.

Thus the algebraic Debye/Stern relation does not replace the Butler–Volmer boundary condition. It only replaces, or initially approximates, the thin electrostatic layer that maps outer electrolyte values to reaction-plane values. BV remains the reactive flux law.

This distinction is important. As an initial condition, the analytical Debye profile cannot change the physical solution if the full solver converges; it only improves basin access. As a reduced boundary model, it becomes a modeling approximation and must be validated against resolved PNP–BV wherever the resolved solver is available.

12 Implementation recommendation

The lowest-risk next step is an optional `debye_boltzmann` initializer for the log-c stack:

1. Use the current mesh coordinate y, nondimensional scaling, BV reaction config, and Boltzmann counterion config.
2. Start with the no-Stern formula above and ignore steric in the IC.
3. Floor O_s, P_s, h_o, H_s only before logarithms and report any floor hit, since that identifies a diffusion-limit regime.
4. Test first near and above +0.60 V, where the current linear IC is far from the Boltzmann layer.
5. Compare against already converged voltages by measuring SNES iterations and final observables. The IC should not change the final converged solution; it should only improve basin access.

13 Answers to the motivating questions

Can the whole system be solved analytically? No. Not for the full two-dimensional production PNP–BV model with finite reaction fluxes, steric terms, and two coupled nonlinear BV reactions.

Can part of the system be solved analytically? Yes. The zero-flux charged layer has a Boltzmann form; the symmetric one-dimensional Debye layer has a Gouy–Chapman solution; the outer electroneutral proton equation reduces to ambipolar diffusion; and the one-dimensional steady outer reaction-diffusion problem reduces to a two-rate algebraic system.

Can we get a good analytical IC for a given voltage? Yes. The matched Debye-layer plus outer algebraic profile is likely the highest-value analytical simplification for the current solver. It directly targets the failure mode documented in `docs/peroxide_window_investigation.md`: Newton starts too far from the exponentially depleted proton layer and the exponentially enriched counterion layer. The proposed IC starts close to that manifold.